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↑ Personal Data:

- Surname: Jamali
- Name: Ramin
- Date and Place of Birth: November 21, 1995; Tehran, Iran
- Nationality: Iranian
- Email: raminjamali@znu.ac.ir,
raminjamali.a@gmail.com
- Web: raminjamali.github.io
- Google Scholar: <https://scholar.google.com/citations?user=H8q0SdoAAAAJ&hl=en>
- Address: Department of Physics, University of Zanjan, Zanjan, Iran
- Tel: +98 911 227 9809

↑ Summary:

- I hold a Ph.D. in Optics and Photonics from the Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran, specializing in optical manipulation, holographic optical tweezers, speckle tweezers, microfluidics, and three-dimensional optical imaging. My research focuses on engineering structured and polarization-controlled optical fields for precise three-dimensional trapping, manipulation, and characterization of micro- and nano-particles. I have developed and experimentally implemented optical systems based on holographic optical tweezers, digital holographic microscopy, and speckle fields for three-dimensional particle manipulation, tracking, and quantitative imaging. My work further encompasses speckle-field engineering, dynamic speckle analysis, optical force measurements, and microfluidic manipulation, with applications ranging from colloidal systems to biological microorganisms. This research combines experimental optics, optical instrumentation, microfluidic platforms, computational imaging, and quantitative data analysis to develop advanced optical approaches for the manipulation, measurement, and three-dimensional characterization of microscopic systems. I completed an INSF postdoctoral position (2025-2026) focused on science promotion and optics education, alongside a visiting lecturer appointment at the University of Zanjan. With over 20 peer-reviewed journal publications, including articles in Nanophotonics, Biomedical Optics Express, Optics Express, and Scientific Reports, Applied Surface Science Advances, and one of which was selected for the Cover Front of Macromolecular Materials and Engineering (December 2024), and three “Image of the Week” recognitions in Optics and Photonics News (Optica), I combine strong experimental optics skills, hands-on instrumentation and device-development experience, and quantitative data analysis (MATLAB, Python, ImageJ) with a track record of interdisciplinary collaboration. My research contributions have been recognized through consecutive Best Ph.D. Student-Researcher awards from Zanjan Province (2023, 2024) and multiple grants from the National Elites Foundation of Iran.

↑ Research Interests:

- Optical Trapping and Manipulation; Speckle Tweezers; Polarization Imaging; Dynamic Speckle Pattern Analysis; Digital Holographic Microscopy (DHM); 3D Imaging and Microscopy; Structured Light Generation; Microfluidics; Depletion and Colloidal Forces; Bacterial Motility Control; Soft Matter and Biophysics; Biomedical Imaging Instrumentation.

↑ Education:

- Ph.D. in Optics and Photonics, 2021–2025
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
Thesis title: Polarization Speckle Tweezers and Speckle Trapping at Fluid-Fluid Interfaces
- M.Sc. in Physics, specializing in Optics, 2018–2021
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
Thesis title: Investigation and Controlling of Colloidal Mixture by Speckle Tweezers

- B.Sc. in Physics, 2014–2018
University of Zanjan, Zanjan, Iran
- Clinical Imaging Training Course
Iran University of Medical Sciences (IUMS), Tehran, Iran
Focus: Principles and clinical practice of medical imaging modalities, bridging quantitative optical/biophysical methods with clinical diagnostic imaging workflows.

↑ Academic and Research Appointments:

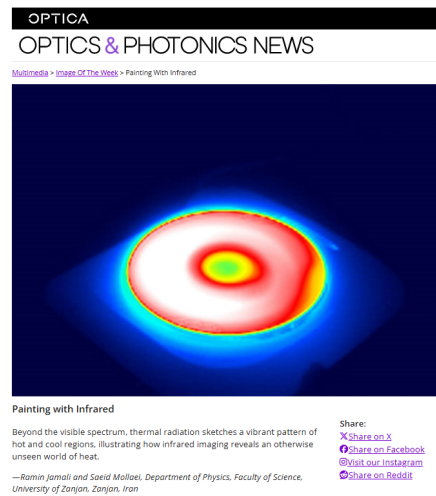
- Invited Lecturer, 2025–2026
Department of Physics, University of Zanjan, Zanjan, Iran
Courses: Physics I (Mechanics), Physics II (Electricity and Magnetism), Physics Laboratory I (Mechanics) , Physics Laboratory II (Electricity and Magnetism)
- Postdoctoral Researcher, Iran National Science Foundation (INSF), 2025–2026
University of Zanjan, Zanjan, Iran
Focus: Science promotion and public engagement activities, together with optics education and training initiatives.
- Selected for the National Elites Foundation Postdoctoral Research Program, 2025

↑ Professional and Industrial Experience:

- Device Developer / R&D Engineer, 2020–2024
Forouzan Zist Fanavaran Co. (knowledge-based technology enterprise), Science and Technology Park of IASBS, Zanjan, Iran
Role: Actively contributed as a developer and manufacturer in the design, prototyping, and fabrication of biological mini-incubator systems and three-dimensional imaging devices. Responsibilities included instrumentation design, optical and mechanical integration, testing and calibration of imaging modules, and troubleshooting of device performance in close collaboration with an interdisciplinary engineering team — experience directly transferable to field-based technical support, system acceptance testing, and instrumentation service for advanced imaging platforms.

↑ Awards and Honors:

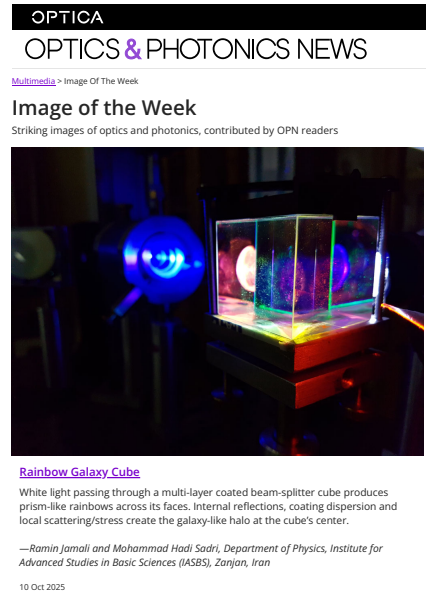
- Selected as the September 2026 “Image of the Week” in Optics and Photonics News (OPN), published by Optica (USA).



- Selected as the May 2026 “Image of the Week” in Optics and Photonics News (OPN), published by Optica (USA).



- Selected as the October 2025 “Image of the Week” in Optics and Photonics News (OPN), published by Optica (USA).



- Selected Ph.D. student of Elites Foundation (individual special project, Zanjan Elites Foundation), 2025
- Best annual Ph.D. student-researcher award, Zanjan Province, 2024
- Best annual Ph.D. student-researcher award, Zanjan Province, 2023
- Selected Ph.D. student of Elites Foundation (group special project, Zanjan Elites Foundation), 2023
- Selected Ph.D. student of Elites Foundation (individual special project, Zanjan Elites Foundation), 2023
- Selected Ph.D. student of Elites Foundation (individual special project, Zanjan Elites Foundation), 2022
- Top student (1st rank, GPA: 19.51/20), Ph.D., IASBS, Zanjan, Iran, 2019
- Top student (1st rank, GPA: 18.33/20), M.Sc., IASBS, Zanjan, Iran, 2019
- Top student (1st rank, GPA: 17.87/20), B.Sc., University of Zanjan, Zanjan, Iran, 2018
- Top student (1st rank, GPA: 17.52/20), High School, Allameh Amini, Zanjan, Iran, 2014
- Top student (1st rank, GPA: 19.90/20), Guidance School, Kharazmi, Zanjan, Iran, 2010
- Top student (1st rank, GPA: 20/20), Elementary School, Ehsan, Zanjan, Iran, 2007

↑ Computer and Programming Skills:

- Scientific Software: MATLAB, $\text{\LaTeX}$, COMSOL Multiphysics, LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator)
- Programming Languages: Python, C++
- Image Analysis: ImageJ, Adobe Photoshop, Adobe InDesign
- 3D and Graphic Design: 3ds Max, Google SketchUp
- Electronic Design: PSpice, ProfiCAD
- Operating Systems: Linux (Ubuntu), Windows (Microsoft Office)

↑ Teaching Experiences:

- Visiting Lecturer, Physics II (Electricity and Magnetism), 2025–2026
University of Zanjan, Zanjan, Iran
- Visiting Lecturer, Physics Laboratory II (Electricity and Magnetism), 2025–2026
University of Zanjan, Zanjan, Iran
- Visiting Lecturer, Physics I (Mechanics), 2025–2026
University of Zanjan, Zanjan, Iran
- Visiting Lecturer, Physics Laboratory I (Mechanics), 2025–2026
University of Zanjan, Zanjan, Iran
- Teaching Assistant, Optics and Photonics I, January 2024–2025
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Modern Physics Laboratory, January 2024–2025
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Optics and Photonics I, January 2023–2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Optics and Photonics II, January 2023–2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Electromagnetism, January 2023–2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Electronics Laboratory, January 2023–2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Electronics, January 2023–2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran

- Teaching Assistant, Modern Physics Laboratory, January 2023–2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Optics and Photonics, January 2021–2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Modern Physics Laboratory, April 2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Electronics, January 2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Electronics Laboratory, January 2018–2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant, Electronics Laboratory, January 2015–2018
University of Zanjan, Zanjan, Iran
- Teaching Assistant, Electromagnetics, February 2017–2018
University of Zanjan, Zanjan, Iran
- Teaching Assistant, Electronics, October 2017–2018
University of Zanjan, Zanjan, Iran
- Teaching Assistant, Optics and Laser, October 2017–2018
University of Zanjan, Zanjan, Iran
- Teaching Assistant, Physics II (Electromagnetics), February 2016–2017
University of Zanjan, Zanjan, Iran

↑ Current Students:

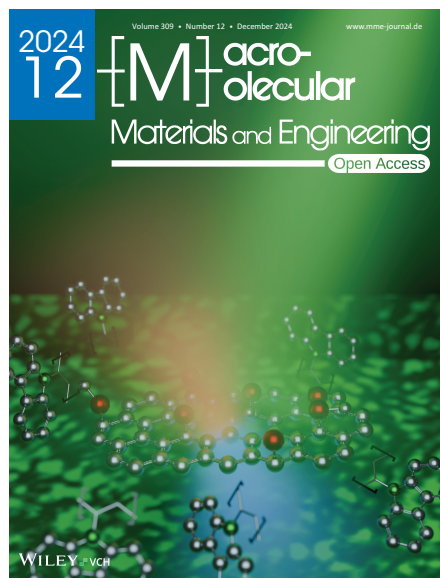
- Ebrahim Nozaripak (Ph.D.), University of Zanjan
- Shima Gholizadeh (Ph.D.), University of Zanjan

↑ Collaboration Research Projects:

1. Investigation of solid-state phenomena using speckle pattern analysis
 - Collaboration with Dr. Mojtaba Nasiri, University of Zanjan, Zanjan, Iran.
2. Investigation of atmospheric and meteorological phenomena using speckle pattern analysis
 - Collaboration with Dr. Ali Bayat, University of Zanjan, Zanjan, Iran.
3. Optical investigation and characterization of liquid crystals
 - Collaboration with Dr. Sergey Shvetsov, on the structural and optical properties of liquid crystalline materials, Yerevan State University.
4. Characterization of industrial engineering chemicals by optical methods
 - Collaboration with Dr. Amin Babaei-Ghazvini, St. Peter's College, Saskatchewan, Canada, on optical characterization of industrial chemical compounds and materials.
5. Real-time characterization and monitoring of lithium battery gel polymers using optical and speckle approaches
 - Collaboration with Dr. Zahra Mohammadnia, Department of Chemistry, IASBS, Zanjan, Iran, on in situ optical and dynamic speckle analysis of gel polymer electrolytes for lithium battery applications.
6. Controlling collective motion by speckle tweezers
 - Collaboration with Dr. Ali-Reza Moradi and Dr. Sabareesh K.P. Velu's group (Rathinam College of Arts and Science, India).
7. 3D imaging using scanning diffractometry with speckle pattern analysis
 - Collaboration with Dr. Ali-Reza Moradi.
8. Metamaterial-assisted illumination nanoscopy via random super-resolution speckles
 - Collaboration with Dr. Ali-Reza Moradi.
9. Phase separation detection of binary mixture by speckle pattern analysis
 - Collaboration with Dr. Ali-Reza Moradi and Dr. Sabareesh K.P. Velu.

↑ Publications — Journals:

20. R. Jamali, Y. Shariati, E. Nozaripak, A. Bayat, M. Sajjadi, M. Gholinejad, F. Kazemi, A. R. Moradi “In situ characterization of Cu-TiO₂ synthesis using dynamic speckle pattern analysis,” Under review in *iScience*, 2026.
19. R. Jamali, M.H. Sadri, and A.R. Moradi, “Remote characterization of aerogel foam concrete using dynamic speckle pattern analysis,” Under review in *Cement and Concrete Research*, 2026.
18. R. Jamali, M.H. Sadri, and A.R. Moradi, “Speckle tweezers in action: manipulating run-and-tumble in bacteria,” Under review in *Optics Letters*, 2026.
17. E. Nozaripak, A. Bayat, R. Jamali, “Multimodal optical characterization of ambient particulate matter-loaded filters using speckle pattern analysis,” *Scientific Reports*, <https://doi.org/10.1038/s41598-026-73636-y>, 2026.
16. M. Sajjadi, R. Jamali, and A.R. Moradi, “Real-time 3D monitoring of NIR laser effects on biodegradable polymers,” *Scientific Reports*, 16, 20805, 2026.
15. E. Nazari, R. Jamali, A. Heidari, Z. Mohamadnia, E. Ahmadi, M. Norouzi, and A.R. Moradi, “Real-time and remote isomerization measurement of photochromic polymer nanoparticles via dynamic speckle pattern analysis,” *The Journal of Physical Chemistry C*, 130, 5559–5573, 2026.
14. R. Jamali, S.K.P. Velu, and A.R. Moradi, “Speckle tweezers near water-oil and water-air interfaces,” *Biomedical Optics Express*, 17(2), 572–599, 2026.
13. M.H. Sadri, R. Jamali, A.J. Khan, F. Rehman, and A.R. Moradi, “Sorting of mesoporous silica derivatives by random optical fields,” *iScience*, 29, 114675, 2026.
12. M.H. Sadri, R. Jamali, and A.R. Moradi, “Speckle fields for controlling and analysis of microscopic collective motion,” *Optics Express*, 33(11), 22864–22882, 2025.
11. M. Sajjadi, R. Jamali, T. Kiyani, Z. Mohamadnia, and A.R. Moradi, “Characterization of Schiff base self-healing hydrogels by dynamic speckle pattern analysis,” *Scientific Reports*, 14(1), 27950, 2024.
10. R. Jamali, M. Sajjadi, B. Taherkhani, D. Abbaszadeh, and A.R. Moradi, “Front Cover: Speckle pattern analysis of PVK-graphene memristor,” *Macromolecular Materials and Engineering*, December 2024.
9. R. Jamali, M. Sajjadi, B. Taherkhani, D. Abbaszadeh, and A.R. Moradi, “Speckle pattern analysis of PVK-graphene memristor,” *Macromolecular Materials and Engineering*, 6:2400213, 2024.
8. R. Jamali, V. Farzam Rad, M. Razaghi, Z. Mohamadnia, M. Khorasani, and A.R. Moradi, “Digital holographic microscopy of spiropyran-based dynamic materials,” *Journal of Microscopy*, 292(2), 2023.



7. R. Jamali, A. Babaei-Ghazvini, E. Nazari, M. Panahi, I. Shahabi-Ghahfarrokhi, and A.R. Moradi, "Surface characterization of biodegradable nanocomposites by dynamic speckle analysis," *Applied Surface Science Advances*, 16, 100429, 2023.
6. O. Pedram, R. Jamali, V. Farzam Rad, R. Khamedi, E. Poursaeidi, and A.R. Moradi, "Evaluation of pitting corrosion by dynamic speckle pattern analysis," *Scientific Reports*, 13(1):8549, 2023.
5. M. Panahi, V. Farzam Rad, Sh. Sasan, R. Jamali, A.R. Moradi, and A. Darudi, "Detection of intralayer alignment in multi-component lipids by dynamic speckle pattern analysis," *Journal of Biophotonics*, <https://doi.org/10.1002/jbio.202200034>, 2022.
4. R. Jamali, F. Nazari, A. Ghaffari, S.K.P. Velu, and A.R. Moradi, "Speckle tweezers for manipulation of high and low refractive index micro-particles and nano-particle loaded vesicles," *Nanophotonics*, 10(11), 2915–2928, 2021.
3. M. Panahi, R. Jamali, V. Farzam Rad, M. Khorasani, A. Darudi, and A.R. Moradi, "3D monitoring of the surface slippage effect on micro-particle sedimentation by digital holographic microscopy," *Scientific Reports*, 11, 12916, 2021.
2. V. Farzam Rad, A. Babaei-Ghazvini, R. Jamali, I. Shahabi-Ghahfarrokhi, and A.R. Moradi, "Digital holographic microscopy for real-time investigation of 3D microstructural dynamics of starch-kefir based nanocomposite," *Applied Optics*, 60(16), 4706–4715, 2021.
1. V. Farzam Rad, M. Panahi, R. Jamali, A. Darudi, and A.R. Moradi, "Non-invasive in situ monitoring of bone scaffold activity by speckle pattern analysis," *Biomedical Optics Express*, 11(11), 6324–6336, 2020.

↑ Publications — Proceedings — International:

6. V. Abbasian, R. Jamali, V. Farzam Rad, A.R. Moradi, and A. Darafsheh, “Dynamic speckle pattern analysis of pitting corrosion,” *Photonic Instrumentation Engineering XI*, 12893, 238–241, March 2024.
5. M.H. Sadri, R. Jamali, F. Rehman, and A.R. Moradi, “Mesoporous material sorting in random energy landscapes,” *11th International Symposium on Optics and its Applications (OPTICS11)*, 89, July 2023.
4. R. Jamali, B. Olamaei, M. Soltanloo, S.E.H. Yeganeh, V. Farzam Rad, J. Esmkhani, and A.R. Moradi, “Measurement of corrosion in power plant turbine rotors by digital holographic microscopy,” *Imaging Congress, hybrid*, 14–17 August 2023.
3. E. Nasimdoust, R. Jamali, F. Amarloo, S.K.P. Velu, and A.R. Moradi, “Controlling spatial distribution of speckle fields for multi-manipulation of colloids,” *Optical Trapping and Optical Micromanipulation XIX*, 21–24 August 2022.
2. R. Jamali, F. Nazari, A. Ghaffari, S.K.P. Velu, and A.R. Moradi, “Simultaneous manipulation of low and high refractive index micro-particles by speckle tweezers,” *PSL Summer School on Soft and Living Matter*, July 2021.
1. R. Jamali, F. Nazari, A. Ghaffari, S.K.P. Velu, and A.R. Moradi, “Collective manipulation of low and high refractive index micro-particles by speckle tweezers,” *International Virtual Conference on Novel Nanomaterials for Innovative Research (ICNNIR)*, April 2021.

↑ Publications — Proceedings — National:

7. M.H. Sadri, R. Jamali, F. Rehman, and A.R. Moradi, “Mesoporous silica sorting by speckle tweezers,” Proc. 29th Iranian Conference on Optics and Photonics and 15th Iranian Conference on Engineering Photonics (ICOP29), February 2023.
6. R. Jamali and A.R. Moradi, “Experimental study of liposome aggregation,” Proc. 26th Annual IASBS Meeting on Condensed Matter Physics, July 2021.
5. R. Jamali, T. Ohadi, and A.R. Moradi, “Investigating the time evolution of color distribution in a variety of surfaces,” Proc. 26th Annual IASBS Meeting on Condensed Matter Physics, July 2021.
4. M. Karimi, R. Jamali, A. Bayat, Y. Abedini, Y. Ghaniun, “Building one channel sun-photometer using light emitting diode,” Proc. 26th Iranian Conference on Optics and Photonics and 12th Iranian Conference on Engineering Photonics (ICOP19), February 2020.
3. R. Jamali and S. Mollaei, “Time evolution of size distribution for oil droplets on water,” Proc. 25th Annual IASBS Meeting on Condensed Matter Physics and School on Topological Phases of Matter, June 2019.
2. A. Bayat, R. Jamali, and S. Mollaei, “Calculation of rotational inertia of gyroscope and measurement of force vector by precession,” Proc. 17th Zanzan Education Research Journal, July 2018.
1. S. Mollaei, R. Jamali, and A.H. Daroneh, “Investigation of self-assembly of aluminium flakes on water surface,” Proc. 24th Annual IASBS Meeting on Condensed Matter Physics and School on Topological Phases of Matter, June 2018.

↑ Talks:

- “Speckle Optical Manipulation and Applications,” Physics Group Seminars, University of Tehran, Tehran, Iran, May 2026.
- “Speckle Tweezers,” Physics Group Seminars, University of Zanjan, Zanjan, Iran, November 2025.
- “Speckle Tweezers at Fluid-Fluid Interfaces,” Optics Group Seminars, IASBS, Zanjan, Iran, February 2024.
- “Speckle Tweezers,” NanoQIQO School on Optics and Photonics (SOP 2023), Yerevan, Armenia, May 2023.
- “Investigation and Controlling of Colloidal Mixture by Speckle Tweezers,” Optics Group Seminars, IASBS, Zanjan, Iran, June 2021.
- “Designing and Construction of a LED Sun Photometer,” University of Zanjan, Zanjan, Iran, June 2017.
- “Calculation of Rotational Inertia of Gyroscope and Measurement of Force Vector by Precession,” University of Zanjan, Zanjan, Iran, December 2016.

↑ Schools and Conferences Attended:

13. NanoQIQO School on Optics and Photonics (SOP), Yerevan, Armenia, May 15–20, 2023.
12. PSL Summer School on Soft and Living Matter, Paris, France, July 5–9, 2021.
11. 26th Annual IASBS Meeting on Condensed Matter Physics, IASBS, Zanjan, Iran, July 7–9, 2021.
10. RSC Desktop Seminar Lectureship on Soft Matter, RSC Webinars, April 29, 2021.
9. International Virtual Conference on Novel Nanomaterials for Innovative Research (ICNNIR), April 29–30, 2021.
8. One-day Online Workshop on Novel Features and Applications of Optical Manipulation, IPM, Tehran, Iran, September 8, 2020.
7. 26th Iranian Conference on Optics and Photonics and 12th Iranian Conference on Engineering Photonics, Kharazmi University (ICOP19), February 2020.
6. 25th Winter School of Physics, IASBS, Zanjan, Iran, February 2020.
5. 25th Annual IASBS Meeting on Condensed Matter Physics, IASBS, Zanjan, Iran, June 13–14, 2019.

4. Workshop on Non-equilibrium Soft Matter, IPM, Tehran, Iran, April 11, 2019.
3. 24th Annual IASBS Meeting on Condensed Matter Physics, IASBS, Zanzan, Iran, June 13–14, 2018.
2. Center for Research in Climate Change and Global Warming (CRCC), University of Zanzan, Zanzan, Iran, December 2015.
1. First Workshop on the Regional Climate Model System (RegCM), December 2015.

↑ Language Skills:

- Mother Tongue (Bilingual): Persian and Azeri
- English:
 - Understanding — Listening: C1 Reading: C1
 - Speaking — Spoken Interaction: B1 Spoken Production: B1
 - Writing: B2

Levels: A1/A2: Basic user B1/B2: Independent user C1/C2: Proficient user

- Good communication skills: active listening, clarifying, and summarising.
- Good command of quantitative data processing and electronic circuit design.

↑ Further Activities:

- Persian Calligraphy
- Watching Movies & Series
- Graphic Design with 3ds Max

↑ Referees:

Dr. Amin Babaei-Ghazvini

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E-mail: amin.babaei@usask.ca, aminb8382@gmail.com

<https://stpeterscollege.ca/faculty/amin-babaeighazvini-2>

Dr. Ali Bayat

Department of Physics, University of Zanjan

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Dr. Zahra Mohammadnia

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